Chapter 1 - Introduction

What is materials science **and engineering** ?

- Materials science involves investigating the relationships that exist between the structures and properties of materials.
- On the other hand, Materials Engineering is, designing or engineering the structure of a material to produce a predetermined set of properties.

- Why should we know about it?

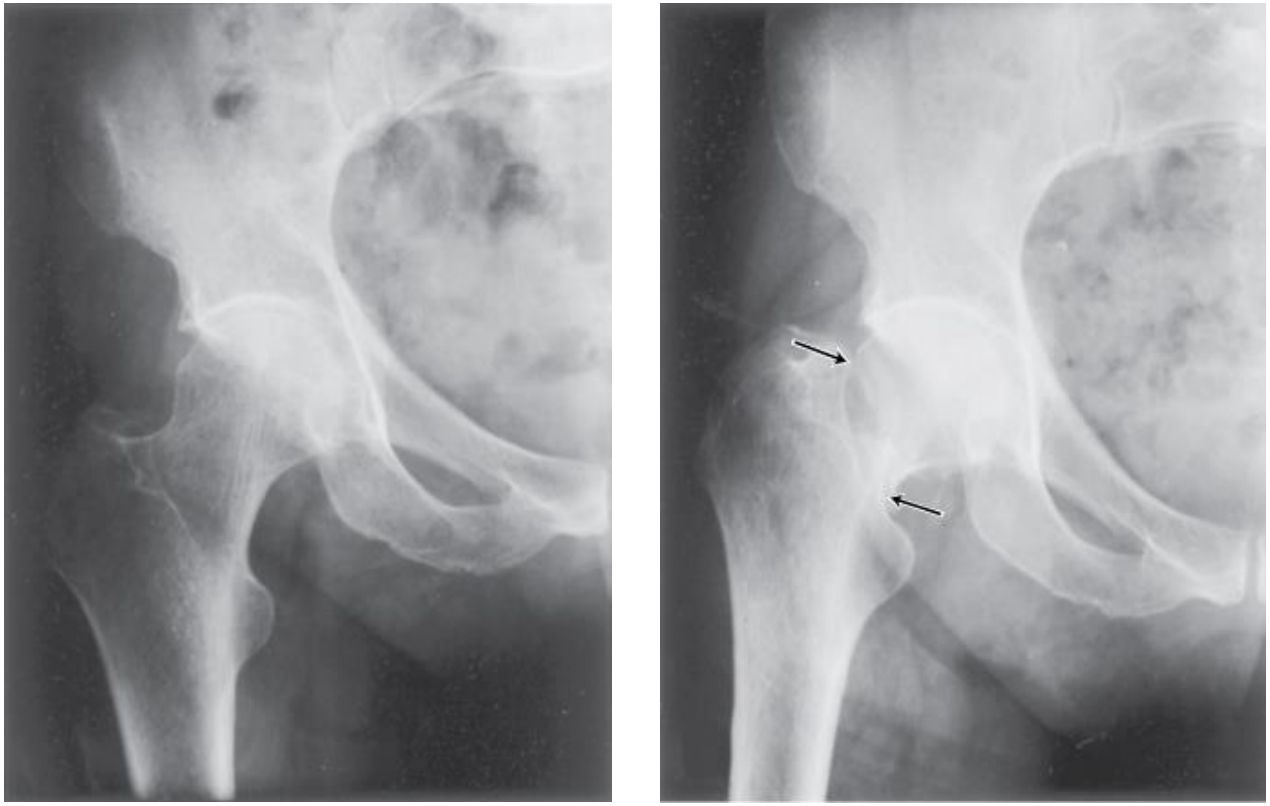
Historical Perspective

- Early civilizations have been designated by the level of their materials development

- Materials drive our society
  - Stone Age
  - Bronze Age
  - Iron Age
  - Now?
    - Silicon Age?
    - Polymer Age?

Example – Hip Implant

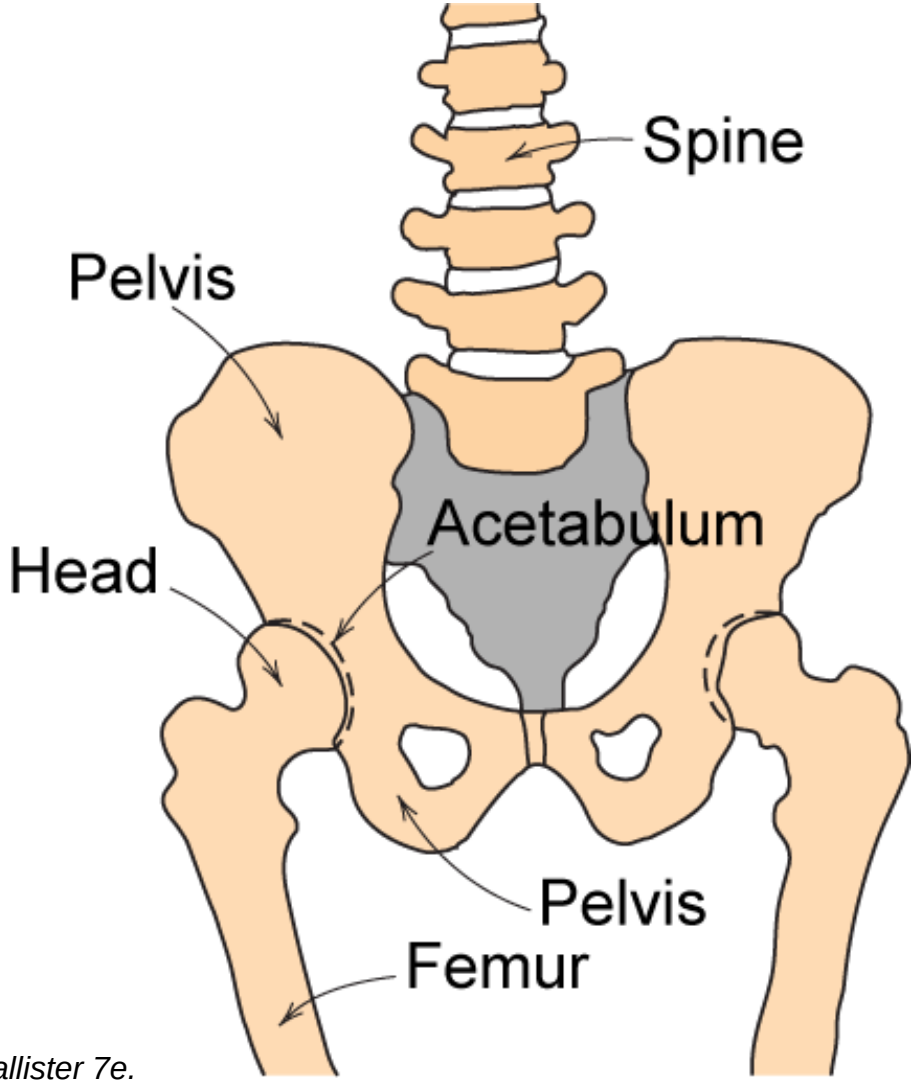
- With age or certain illnesses joints deteriorate. Particularly those with large loads (such as hip).



Adapted from Fig. 22.25, *Callister 7e.*

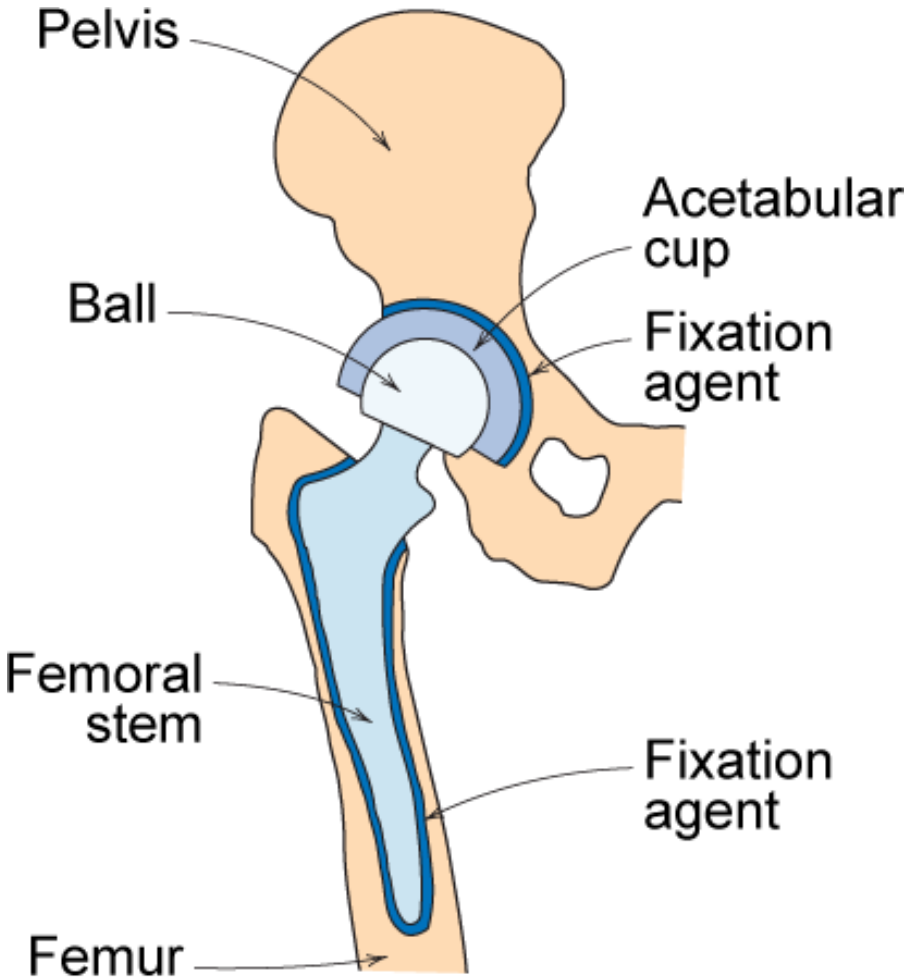
Example – Hip Implant

- Requirements
  - mechanical strength (many cycles)
  - good lubricity
  - biocompatibility



Adapted from Fig. 22.24, *Callister 7e.*

Example – Hip Implant



(a)

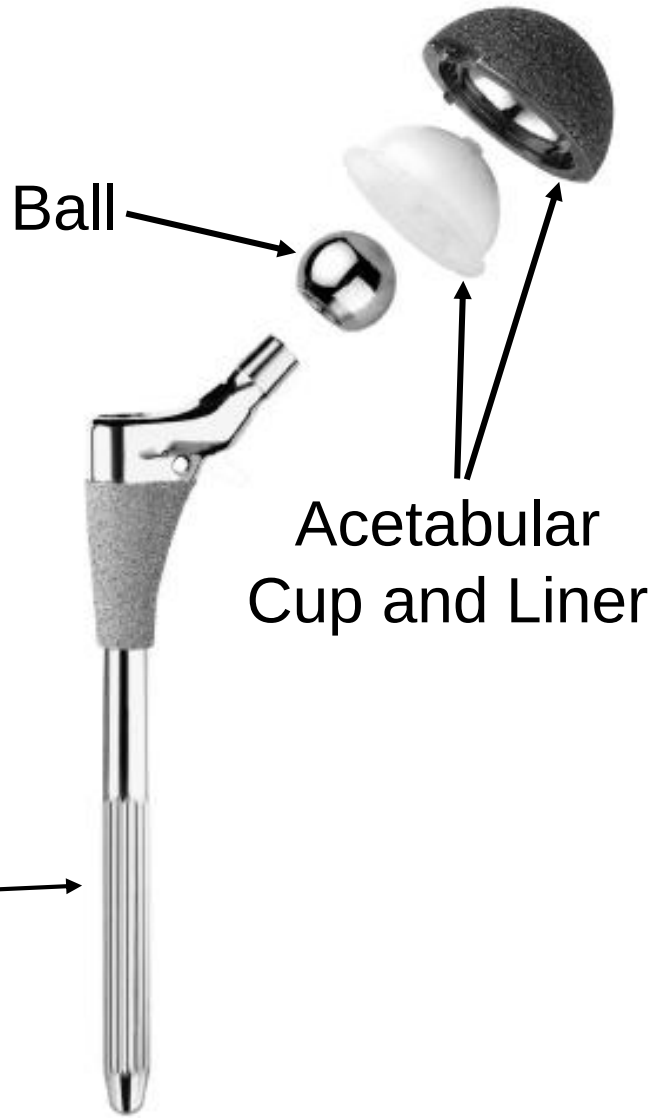


(b)

Adapted from Fig. 22.26, *Callister 7e.*

Hip Implant

- Key problems to overcome
  - fixation agent to hold acetabular cup
  - cup lubrication material
  - femoral stem – fixing agent
  - must avoid any debris in cup



Adapted from chapter-opening photograph, Chapter 22, *Callister 7e.* (Photograph courtesy of Zimmer, Inc., Warsaw, IN, USA.)

Structure, Processing, & Properties

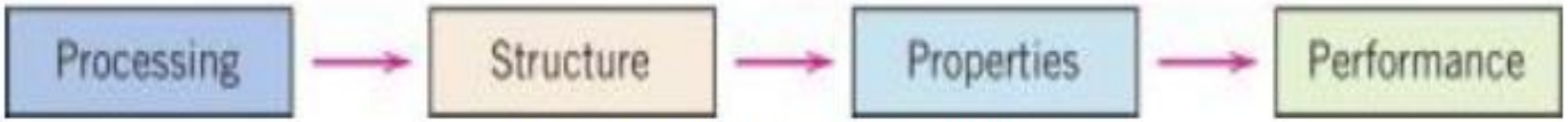
- Structure: relates to the arrangement of a material's internal components
  - Microscopic
  - Macroscopic
- Properties: a material trait in terms of kind and magnitude of response to a specific imposed stimulus. Six different categories
  - Mechanical, stimulus is applied load
  - Electrical, stimulus is electric field
  - Thermal, temperature

Structure, Processing, & Properties

- Magnetic, magnetic field
- Optical, electromagnetic or light radiation
- Deteriorative, chemical reactivity

• Processing: the structure of a material will depend on how it is processed.

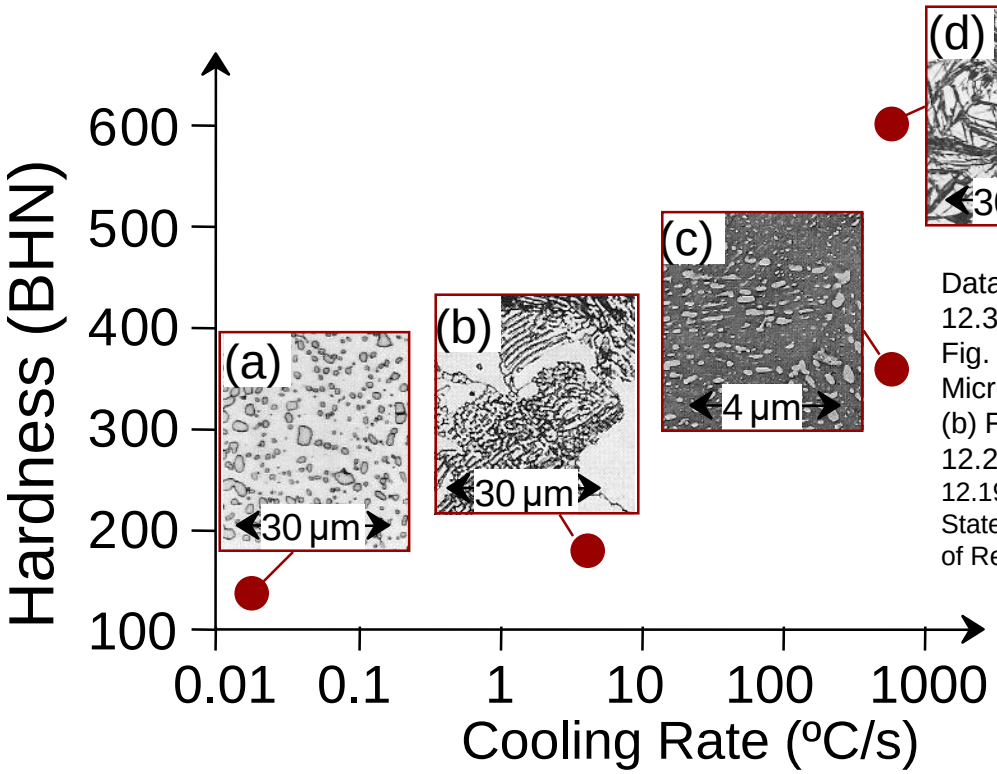
- The interrelationship between processing, structure, properties, and performance is depicted in the schematic illustration



Adapted from Chapter 1, *Callister 8e*

Structure, Processing, & Properties

- Properties depend on structure
  ex: hardness vs structure of steel



Data obtained from Figs. 12.31(a) and 12.32 with 4 wt% C composition, and from Fig. 17.8, *Callister & Rethwisch 9e*. Micrographs adapted from (a) Fig. 12.19; (b) Fig. 11.29; (c) Fig. 12.33; and (d) Fig. 12.21, *Callister & Rethwisch 9e*. (Figures 12.19, 12.21, & 12.33 copyright 1971 by United States Steel Corporation. Figure 9.30 courtesy of Republic Steel Corporation.)

- Processing can change structure
  ex: structure vs cooling rate of steel

Example – Structure, Processing, & Properties



Three thin disk specimens of aluminium oxide placed over a printed page

Types of Materials

- Metals:
  - Strong, ductile
  - High thermal & electrical conductivity
  - Opaque, reflective.

- Polymers/plastics: Covalent bonding → sharing of electrons
  - Soft, ductile, low strength, low density
  - Thermal & electrical insulators
  - Optically translucent or transparent.

- Ceramics: ionic bonding (refractory) – compounds of metallic & non-metallic elements (oxides, carbides, nitrides, sulfides)
  - Brittle, glassy
  - Non-conducting (insulators)

Types of Materials, cont'd

- **Advanced Materials** used in high-tech applications. These include:

- Semiconductors (having electrical conductivities intermediate between conductors and insulators)
- Biomaterials (which must be compatible with body tissues)
- Smart materials (those that sense and respond to changes in environments in predetermined manners e.g. optical fibers, piezoelectric materials and microelectromechanical systems-MEMS)
- Nanomaterials (those that have structural features on the order of a nanometer, some of which may be designed on the atomic/molecular scale).

Examples – Types of Materials



Metallic materials (silverware, scissors, coins, a gear, a wedding ring, a nut and bolt)

Examples – Types of Materials, cont'd



Ceramic materials (scissors, tea cup, building brick, floor tile, glass vase

The Materials Selection Process

1. Pick Application ⟶ Determine required Properties

   Properties: mechanical, electrical, thermal, magnetic, optical, deteriorative.

2. Properties ⟶ Identify candidate Material(s)

   Material: structure, composition.

3. Material ⟶ Identify required Processing

   Processing: changes *structure* and overall *shape*
   ex: casting, sintering, vapor deposition, doping
   forming, joining, annealing.

ELECTRICAL

- Electrical Resistivity of Copper:

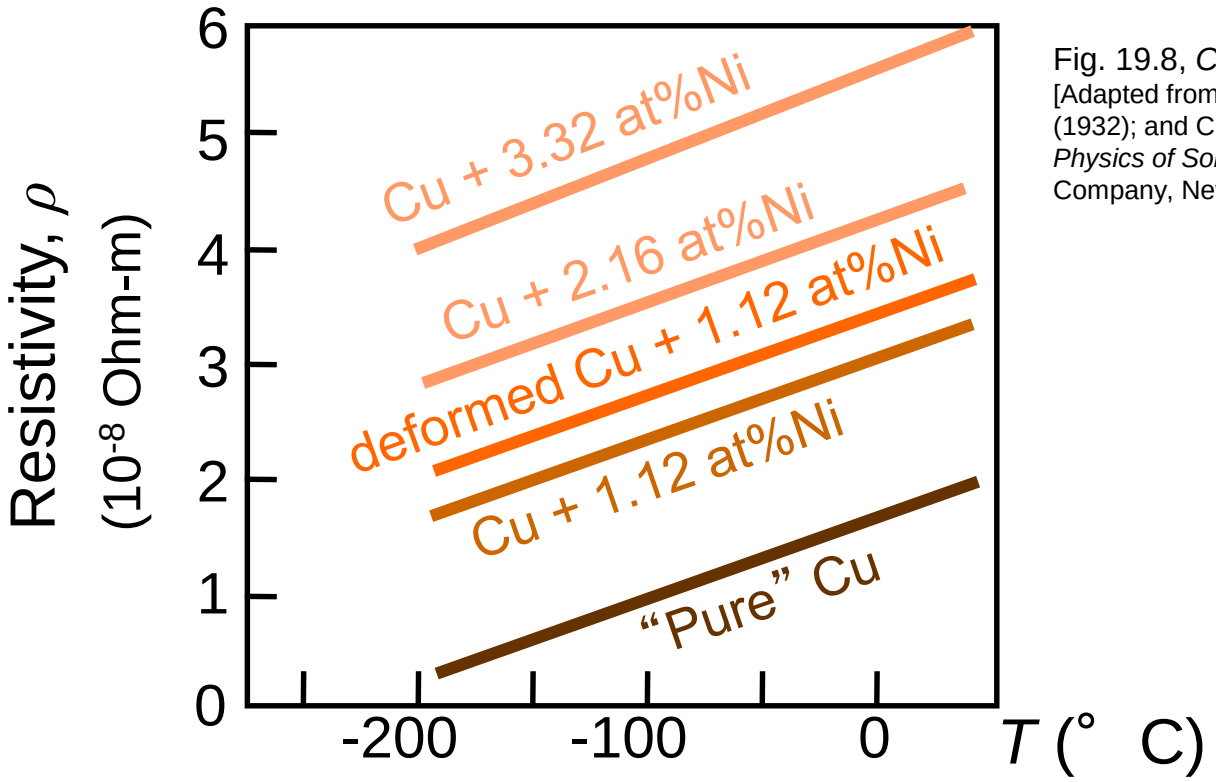


Fig. 19.8, *Callister & Rethwisch 9e.* [Adapted from: J.O. Linde, *Ann Physik* **5**, 219 (1932); and C.A. Wert and R.M. Thomson, *Physics of Solids*, 2nd edition, McGraw-Hill Company, New York, 1970.]

- Adding "impurity" atoms to Cu increases resistivity.
- Deforming Cu increases resistivity.

THERMAL

- Space Shuttle Tiles:
  -- Silica fiber insulation offers low heat conduction.



Chapter-opening photograph, Chapter 17, *Callister & Rethwisch 3e.* (Courtesy of Lockheed Missiles and Space Company, Inc.)

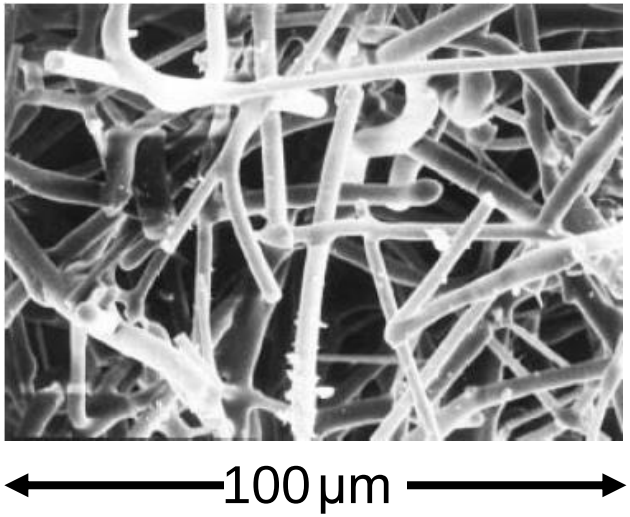


Fig. 19.4W, *Callister 6e.* (Courtesy of Lockheed Aerospace Ceramics Systems, Sunnyvale, CA) (Note: "W" denotes fig. is on CD-ROM.)

- Thermal Conductivity of Copper:
  -- It decreases when you add zinc!

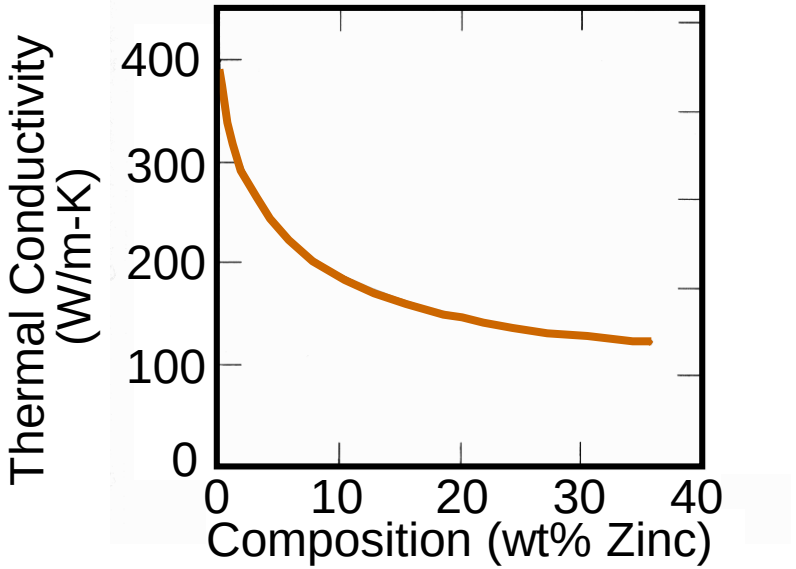


Fig. 20.4, *Callister & Rethwisch 9e.* [Adapted from *Metals Handbook: Properties and Selection: Nonferrous alloys and Pure Metals*, Vol. 2, 9th ed., H. Baker, (Managing Editor), ASM International, 1979, p. 315.]

MAGNETIC

- Magnetic Storage:
  -- Recording medium is magnetized by recording head.

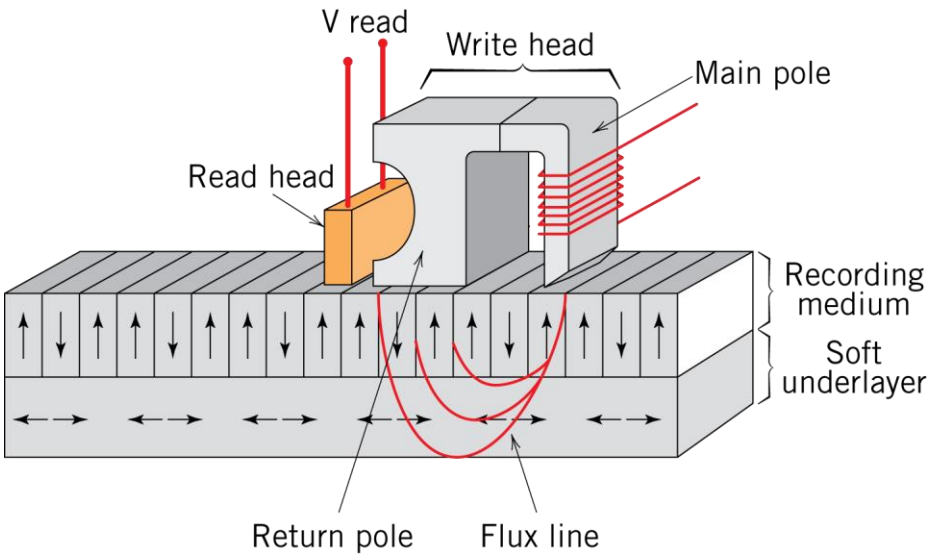
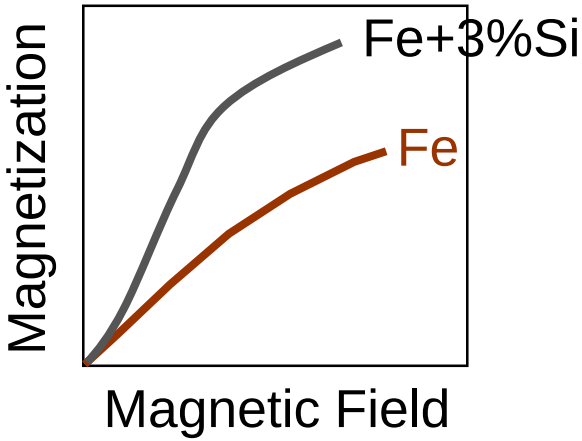


Fig. 21.23, *Callister & Rethwisch 9e.*
(Courtesy of HGST, a Western Digital Company.)

- Magnetic Permeability vs. Composition:
  -- Adding 3 atomic % Si makes Fe a better recording medium!



Adapted from C.R. Barrett, W.D. Nix, and A.S. Tetelman, *The Principles of Engineering Materials*, Fig. 1-7(a), p. 9, 1973. Electronically reproduced by permission of Pearson Education, Inc., Upper Saddle River, New Jersey.

OPTICAL

- Transmittance:
  -- Aluminum oxide may be transparent, translucent, or opaque depending on the material's structure (i.e., single crystal vs. polycrystal, and degree of porosity).

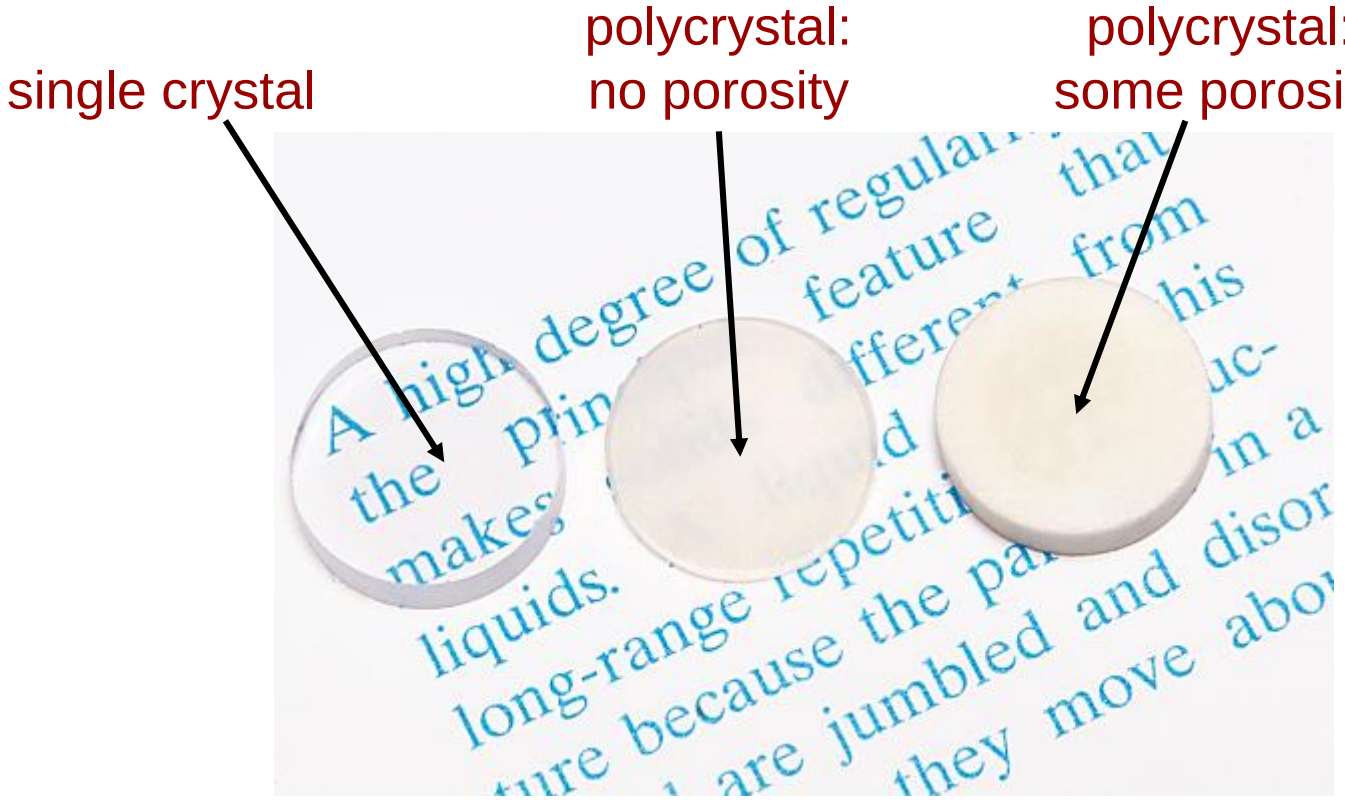


Fig. 1.2, *Callister & Rethwisch 9e.*
(Specimen preparation, P.A. Lessing)

DETERIORATIVE

- Stress & Saltwater...
  -- causes cracks!

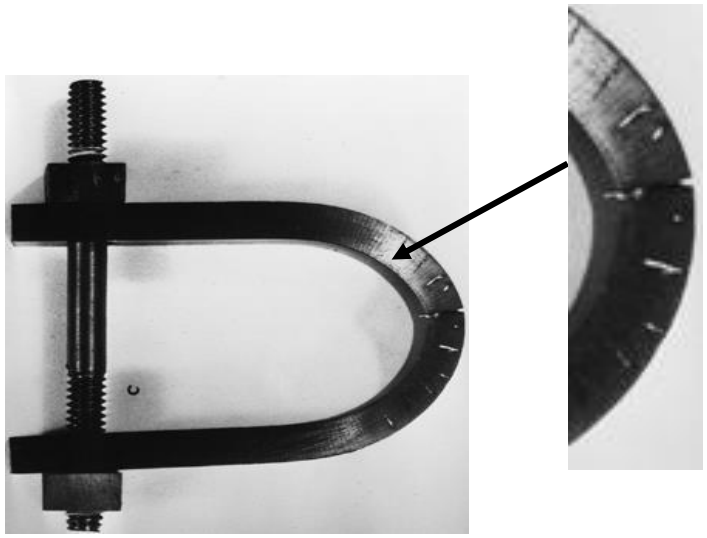
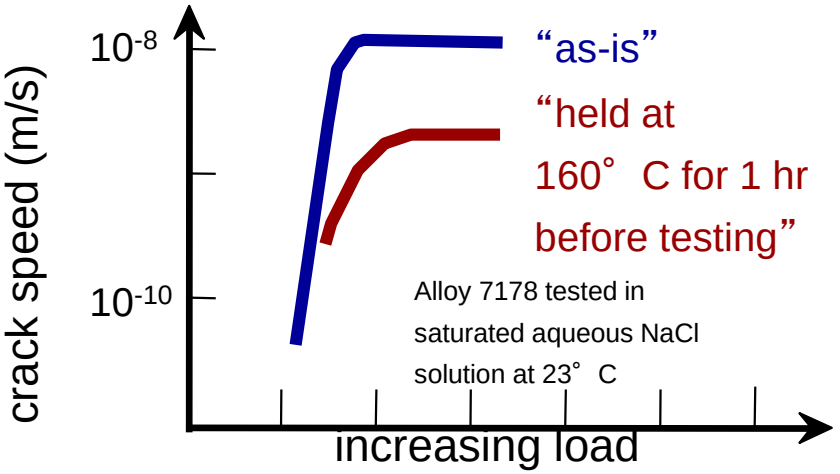


Fig. 18.21, *Callister & Rethwisch 9e.*
(from *Marine Corrosion, Causes, and Prevention*, John Wiley and Sons, Inc., 1975.)

- Heat treatment: slows crack speed in salt water!



Adapted from Fig. 11.20(b), R.W. Hertzberg, "Deformation and Fracture Mechanics of Engineering Materials" (4th ed.), p. 505, John Wiley and Sons, 1996. (Original source: Markus O. Speidel, Brown Boveri Co.)

SUMMARY

Course Goals:

- Use the right material for the job.
- Understand the relation between properties, structure, and processing.
- Recognize new design opportunities offered by materials selection.

ANNOUNCEMENTS

Reading:

Core Problems:

Self-help Problems: